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Wei-Cheng Lee

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Four-Channel Short-Wavelength Division Multiplexing with Sub-Wavelength Metastructures in Silicon Nitride for High-Performance Optical Backplanes

Supervising Committee:

Supervisor: Ray Chen

Cheng Guo

Four-Channel Short-Wavelength Division Multiplexing with Sub-Wavelength Metastructures in Silicon Nitride for High-Performance Optical Backplanes

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Abstract

Four-channel short-wavelength division multiplexing with sub-wavelength metastructures in silicon nitride for high-performance optical backplanes

Wei-Cheng Lee, MSE

The University of Texas at Austin, 2026

Supervisor: Ray Chen

A four-channel short-wavelength-division-multiplexing (SWDM) device is demonstrated through inverse design in a Si_3N_4-SiO_2 integrated photonics platform for operation across the λ = 850–940 nm spectral band with 6x6 μm^2 footprint. Sub-wavelength metastructures are employed to aggregate four wavelength channels within a compact footprint. The metastructure provides broadband modal control and wavelength selectivity while maintaining fabrication tolerance. Finite-difference time-domain (FDTD) simulations show insertion losses between 1.25 and 1.87 dB per channel. The proposed architecture enables multi-wavelength optical backplanes with scalable fan-in capability for high-speed avionics and short-reach data links.

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Chapter 1: Introduction

1.1 Motivation for Optical Backplanes

Electrical interconnects in high-speed backplanes face fundamental limitations that increasingly hinder their scalability and performance in next-generation computing and communication systems. As data rates continue to rise and transmission distances grow, copper-based links become constrained by signal attenuation, reflections, crosstalk, inductive effects, excessive power dissipation, and inherent bandwidth scaling bottlenecks [1][2]. At high frequencies, skin effect and dielectric absorption further degrade signal integrity over longer distances, significantly reducing system efficiency [2].

A variety of advanced techniques have been proposed to extend the performance of electrical interconnects [3][4][5][6]. Equalization, for example, can partially mitigate signal distortion and extend the usable bandwidth of copper links [3]. However, these methods cannot overcome the fundamental scaling limitations of wiring. The intrinsic energy and density bottlenecks remain unresolved, limiting their long-term applicability in ultra-dense backplane environments [6]. Moreover, achieving higher bandwidth with copper requires either the adoption of expensive low-loss dielectric materials or the use of wider trace geometries. Both strategies compromise routing density, increase design complexity, and raise overall system cost [7]. These physical and material constraints have become key barriers to the continued advancement of electrical interconnects.

In contrast, optical interconnects present a compelling solution to these challenges. Optical links are inherently immune to electromagnetic interference (EMI) and electromagnetic pulses (EMP), ensuring robust signal integrity in electrically noisy

environments [8]. Beyond this, optical transmission exhibits significantly lower signal attenuation over distance and supports much higher interconnect density [9][10]. Owing to their compact cross-sectional footprint, optical waveguides enable the simultaneous transmission of multiple channels within the same spatial area, surpassing the scalability of electrical traces. Furthermore, optical interconnects offer superior energy efficiency at high data rates, making them particularly attractive for large-scale systems such as data centers, avionics platforms, and high-performance computing environments.

1.2 Overview of optical backplanes

Optical backplanes have emerged as a promising technology to address the limitations of traditional electrical interconnects in high-performance computing, data centers, and avionics systems. An optical backplane is essentially a communication platform in which optical interconnects replace conventional copper traces to provide high-bandwidth, low-loss, and energy-efficient data transfer between modules and subsystems [11]. By shifting from electrical to optical signaling, backplanes can overcome many of the scalability bottlenecks that restrict next-generation architecture.

From a system-level perspective, optical backplanes can be implemented using different architectural approaches. Two common strategies include bus-based and point-to-point interconnect configurations [12]. Bus-based designs allow multiple modules to share a common optical link, offering simplicity and cost savings, but bus structures often face challenges in managing bandwidth allocation and crosstalk. Point-to-point designs, on the other hand, establish direct optical links between communicating nodes, providing higher

bandwidth and improved isolation at the expense of increased routing and packaging complexity [13]. Alternative network topologies, such as star, mesh, and ring interconnects [14], are also being explored to balance scalability, redundancy, and power efficiency in optical backplane systems.

Although optical backplanes provide significant benefits such as immunity to electromagnetic interference, reduced signal attenuation over distance, higher interconnect density, compact footprint, and superior energy efficiency at high data rates, several challenges remain for widespread adoption. These challenges include the difficulty of integrating optical components with existing electronic systems, the need for high alignment and packaging precision, thermal stability concerns, and overall manufacturing cost [15]. In addition, practical optical backplane implementation requires reliable sources, detectors, and modulators, as well as efficient coupling techniques between fibers, waveguides, and photonic integrated circuits [16][17]. Addressing these challenges is essential to enable scalable and commercially viable optical backplane systems.

1.3 Meta-structure and Guided-wave optics

Optical waveguides are the fundamental building blocks of optical interconnects and backplanes. A waveguide is a dielectric structure that confines and guides light through refractive index contrast between a high-index core and a lower-index cladding. By controlling this confinement, waveguides can support efficient signal transmission with low loss and high integration density [18]. Modern optical backplane designs employ two main categories of waveguide structures: conventional guided-wave geometries and

engineered meta-structures. Conventional guided-wave structures rely on total internal reflection for light confinement, whereas meta-structures utilize sub-wavelength patterning to achieve effective index modulation and novel functionalities beyond those offered by traditional geometries [19]. A clear understanding of these two classes is essential for the design of scalable, high-performance optical interconnects.

Conventional guided-wave optics is based on the principle of total internal reflection, where light is confined within a core material of higher refractive index surrounded by a lower-index cladding. The guided modes can be described by the effective index method, which captures how the electromagnetic field propagates through the confined geometry [18].

Several waveguide geometries are commonly used in integrated photonic systems. Strip and rib waveguides offer strong mode confinement and compatibility with CMOS fabrication processes, making them widely adopted in silicon and silicon nitride photonics [20]. Ridge waveguides provide lower scattering loss and improved fabrication tolerance, while slot waveguides exploit a narrow low-index region to concentrate the optical field, enabling strong light–matter interaction for sensing and modulation applications [20]. Depending on the design, waveguides may operate in single-mode or multimode regimes, with single-mode operation preferred for minimizing dispersion and crosstalk in dense backplane environments [21][22].

Key performance metrics of guided-wave structures include propagation loss, mode confinement factor, dispersion, and bend radius. Low-loss waveguides are particularly critical for backplane applications, where high interconnect density and long routing paths

demand minimal attenuation [23]. In addition, coupling efficiency between waveguides, fibers, and other integrated components strongly impacts system scalability [24]. Thus, conventional guided-wave structures provide the foundation for practical optical backplanes but are often limited by trade-offs in footprint, dispersion control, and polarization dependence.

Meta-structured waveguides represent an advanced class of optical guiding structures in which sub-wavelength features are engineered within the waveguide core or cladding to tailor electromagnetic properties. Instead of relying solely on bulk index contrast, these structures exploit periodic or quasi-periodic modulation to realize effective medium behavior and precise control over the optical mode [25].

One widely adopted approach is the sub-wavelength grating (SWG) waveguide, where alternating regions of high- and low-index material create an effective refractive index that can be tuned by adjusting the duty cycle and pitch of the grating [26]. This enables smooth transitions, low-loss bends, and compact mode converters that outperform conventional designs. Another important category includes photonic crystal waveguides, which employ periodic lattice structures to engineer photonic bandgaps, allowing for dispersion engineering and compact filtering functions [27]. More recently, metasurface-inspired structures have been explored for polarization management, broadband mode conversion, and highly compact wavelength-division multiplexing (WDM) devices [28].

Meta-structured waveguides offer several advantages that are particularly relevant to the development of optical backplanes. One important advantage is the ability to achieve precise dispersion control, which enables broadband and low-loss transmission [28]. In

addition, the high degree of design flexibility provided by meta-structures facilitates device miniaturization and supports higher interconnect densities. When properly optimized, meta-structured designs can also enhance robustness against fabrication variations, thereby improving overall reliability [29]. Despite these benefits, the adoption of meta-structured waveguides introduces challenges, including increased fabrication complexity and heightened sensitivity to sub-wavelength feature accuracy, which may negatively affect device yield and scalability.

1.4 Common wavelengths for optical interconnect and communication

The operating wavelength is a critical parameter in optical backplane design, as it determines propagation loss, dispersion behavior, device compatibility, and overall system performance. Optical interconnects for board-level and chip-to-chip communication have largely converged on a few well-established spectral regions that balance low-loss transmission with the availability of reliable sources and detectors.

In short-reach backplane environments, two wavelength regimes are predominantly used. The 850 nm band, driven by vertical-cavity surface-emitting lasers (VCSELs), has been widely adopted for multimode fiber and polymer waveguide systems because of its low cost, compact sources, and high reliability [30]. The 1310 nm band is another important choice, particularly in silicon photonics platforms, since it provides a favorable trade-off between attenuation, dispersion, and device maturity [31]. In some cases, the 1550 nm band is also employed when compatibility with existing single-mode fiber infrastructure is required for board-to-board interconnects [32].

Extending optical backplane operation into other spectral regions remains challenging. Although alternative wavelengths could in principle offer benefits in dispersion control or integration density, their practical adoption is limited by the lack of reliable light sources, detectors, and modulators, as well as increased propagation losses in many material platforms [33]. Consequently, optical backplane development continues to focus on the 850 nm, 1310 nm, and 1550 nm ranges, where mature device ecosystems and fabrication processes are already established.

1.5 Material platform selection

The choice of material platform is a decisive factor in optical backplane development, as it governs transparency range, propagation loss, fabrication maturity, integration feasibility, and overall cost. Different material systems have been investigated for guided-wave interconnects, each offering unique trade-offs in performance and manufacturability. The most prominent platforms include silicon photonics, silicon nitride, silica on silicon, and III–V compound semiconductors.

Silicon photonics has become a leading platform due to its compatibility with CMOS fabrication processes and its established manufacturing infrastructure [34]. The high refractive index contrast of silicon enables compact waveguides and dense integration. However, crystalline silicon is transparent only above ~1.1 μm, restricting operation to the O-, C-, and L-bands [35]. Moreover, nonlinear loss mechanisms such as two-photon absorption and free-carrier absorption can degrade performance at high optical intensities [36]. Despite these limitations, silicon photonics remains a strong candidate for optical

backplanes where dense integration with electronics is a primary requirement.

Silicon nitride (Si_3N_4) provides a complementary alternative with a broad transparency window from the visible spectrum to beyond 2 μm. Silicon nitride waveguides exhibit exceptionally low propagation loss, in some cases below 0.1 dB/cm in optimized fabrication processes [37]. These properties make Si_3N_4 particularly attractive for short-wavelength division multiplexing (SWDM) in the 850–940 nm range, where silicon is not transparent. However, challenges such as film stress and cracking in thick layers must be carefully managed in practical implementations [37].

Silica on silicon is another widely used platform, particularly for passive photonic circuits. Its primary advantage lies in its ultra-low propagation loss and excellent fiber compatibility, since its refractive index closely matches that of standard optical fibers [38]. This enables highly efficient coupling and high signal integrity across chip–fiber interfaces. However, the low index contrast between core and cladding requires thick waveguide layers and large bending radii, leading to significantly larger device footprints compared to high-index-contrast platforms such as Si or Si_3N_4. Furthermore, achieving ultra-low loss relies on careful material and fabrication control in doped silica waveguide platforms, where optimized composition and processing enable low-loss optical performance [39]. As a result, silica is well suited for low-loss passive backplane interconnects, but less favorable where compactness and integration density are critical.

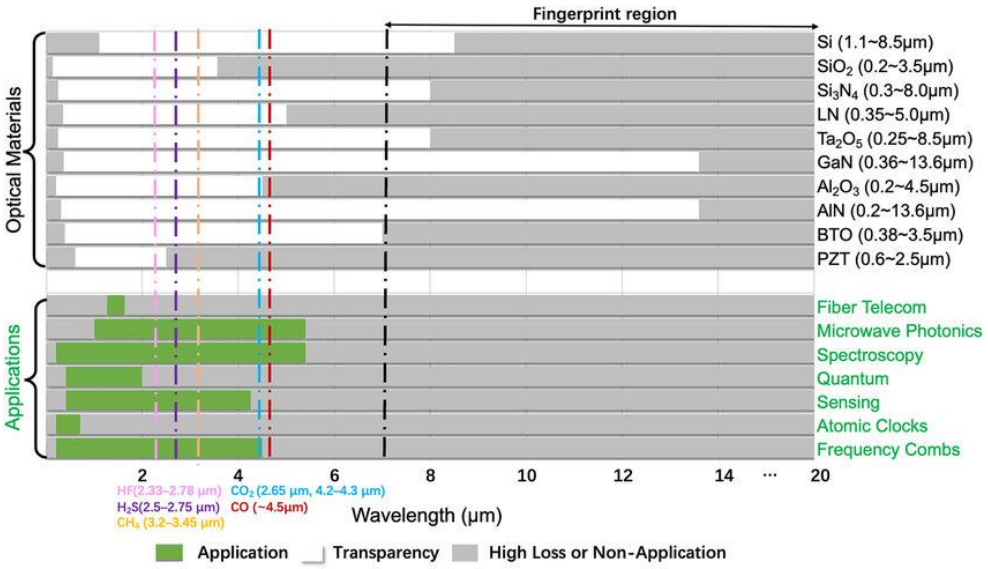


Figure 1.1: Ultra-wideband integrated photonic devices on silicon platform: from visible to mid-IR [35].

III–V compound semiconductors, including indium phosphide (InP) and gallium arsenide (GaAs), offer unique advantages due to their direct bandgap, which enables efficient light emission and detection [40][41]. These materials allow monolithic integration of active devices such as lasers, modulators, and detectors within the same platform, particularly for the C- and L-bands [42]. While their performance is well suited for high-capacity links, the relatively high cost of fabrication and challenges in scaling wafer-level processes limit their application in cost-sensitive board-level interconnects [43].

Chapter 2: Methodology of Inverse Design

2.1 Overview of Inverse Design in Nanophotonics

The design of nanophotonic devices has long been dominated by intuition-driven and analytical approaches [44]. Traditional components such as directional couplers, multimode interferometers (MMIs), and grating couplers are based on well-known physical principles—interference, phase matching, or coupling theory—and are typically optimized through manual parameter sweeps or semi-empirical adjustment [45]. While these strategies have established a large library of reliable designs, they rely on a limited set of parameters and assume highly simplified field interactions [44].

As integrated photonic systems move toward greater functional density, broader bandwidth, and smaller footprints, these conventional methods increasingly fail to achieve the required performance. A typical forward-design process starts with a predefined geometry whose parameters are tuned to meet a performance target. For example, the width and length of an MMI can be scanned to improve the splitting ratio, or the etch depth of a grating coupler can be varied to increase coupling efficiency [46].

Although effective for devices governed by simple modal behavior, this process becomes inefficient or infeasible when multiple objectives—such as minimizing crosstalk, maintaining broadband operation, and reducing loss—must be satisfied simultaneously [47]. The resulting designs are therefore often sub-optimal and template-bounded, limited by the designer's intuition and the geometry's analytic form [48].

In contrast, inverse design adopts an algorithmic and physics-driven strategy: instead of tuning a few geometric parameters, the entire permittivity distribution within the design

region is treated as a variable to be optimized under Maxwell's equations [44]. Given a desired functionality, such as a specific mode conversion, polarization rotation, or spectral response, the algorithm iteratively modifies the spatial dielectric profile to minimize a defined figure of merit (FOM), for example, transmission loss or mode overlap error. The optimization proceeds automatically through closed-loop feedback between electromagnetic simulation and gradient evaluation, continuing until the FOM converges to the optimal value or the iteration reaches the preset limit [48]. This approach removes the constraints of human intuition and allows the discovery of non-intuitive topologies that perform efficiently across multiple operating conditions.

Simple examples highlight the contrast between the two paradigms. In a conventional design [49], a Y-branch splitter is optimized by adjusting the branching angle to achieve balanced power splitting; in inverse design [50], the entire region connecting the input and output waveguides can freely evolve, often producing an irregular but highly efficient topology that achieves the same goal within a footprint several times smaller. Similarly, while an MMI-based mode converter typically requires tens of micrometers to operate effectively [51], topology optimization has produced ultra-compact 6 μm × 6 μm mode multiplexers capable of supporting multiple orthogonal modes over a 100 nm bandwidth—a performance unreachable by parameter-swept MMIs [52]. These examples demonstrate that inverse design not only automates the optimization process but also expands the reachable design space, enabling device performances near theoretical limits.

2.2 Mathematical Formulation of the Inverse Problem

The inverse-design methodology reformulates an optical performance objective into a mathematically constrained optimization problem governed by Maxwell's equations. This translation from physical function to mathematical structure defines how the electromagnetic fields, material parameters, and optimization variables interact within a unified computational framework. A rigorous formulation is essential to ensure that the resulting optimization converges toward physically meaningful, numerically stable, and fabrication-compatible solutions rather than purely numerical artifacts.

The first step in constructing an inverse-design problem is to translate the intended optical function—such as equal power splitting, polarization conversion, or broadband filtering—into a quantitative objective function or figure of merit (FOM) [53]. FOM serves as a mathematical expression of device performance and may represent transmission efficiency, reflection loss, mode-conversion fidelity, or field overlap with a target mode. To accurately represent the physical system, the formulation must include the governing electromagnetic relations, typically expressed through Maxwell's equations, together with appropriate boundary conditions, excitation sources, and material properties that describe the operational environment. These components ensure that the optimization process evaluates performance under conditions consistent with the actual electromagnetic behavior of the device.

Once the objective is defined, the design variables and constraint conditions establish the mathematical degrees of freedom and feasible design space. The design variables are most commonly represented by the spatial distribution of the relative permittivity $\varepsilon(r)$ or

by an equivalent normalized material-density field $\rho(r)$ [56]. A continuous formulation, where $\rho(r)$ varies smoothly between the constituent materials, enables the use of gradient-based optimization algorithms and efficient sensitivity analysis. This approach allows gradual adjustment of optical properties within the design domain while maintaining differentiability of the objective function [53].

The constraints embedded in the formulation are critical to maintaining physical realism and manufacturability. Typical constraint categories include:

- Material bounds: Restricting the permittivity distribution, $\varepsilon(r)$, to the refractive-index range of available materials [53].
- Geometric limitations: Such as minimum feature size or layer thickness, reflecting lithographic and etching capabilities [54].
- Energy-conservation and passivity requirements: Preventing non-physical field amplification [55].
- Boundary conditions: Defining how the fields interact with ports, claddings, or periodic interfaces.

Together, these constraints delimit the permissible design space to physically realizable solutions. An effective mathematical formulation therefore balances flexibility, allowing exploration of complex geometries, with feasibility, ensuring that each iteration of the optimization remains consistent with electromagnetic and fabrication constraints.

A well-posed formulation requires careful consideration of numerical accuracy, robustness, and scaling. The mesh resolution and time-step size in numerical solvers must

capture the relevant field features without causing excessive computational overhead. Proper normalization between multiple objective terms is necessary to prevent imbalance between competing performance targets. Furthermore, robustness to fabrication variation is an essential component of modern inverse-design practice; small perturbations in geometry should not lead to significant degradation in optical performance. To achieve this, stochastic sampling or ensemble-averaged objectives are occasionally incorporated during optimization to maintain consistent behavior under realistic fabrication tolerances.

2.3 Optimization Frameworks for Inverse Design

The mathematical formulation in Section 2.2 defines the objective function, design variables, and physical constraints that characterize the inverse-design problem. To obtain a structure that satisfies this formulation, an optimization procedure must update the design variables in a manner that improves the prescribed figure of merit while maintaining consistency with Maxwell's equations. Effective optimization is therefore central to the inverse-design framework and determines both the convergence behavior and the quality of the final device.

Large-scale photonic inverse design requires an optimization strategy capable of handling thousands to millions of degrees of freedom. Gradient-based methods satisfy this requirement and have become the dominant approach due to their deterministic convergence and computational efficiency. These methods rely on the adjoint-variable technique, which computes the gradient of the figure of merit with respect to the material distribution using one forward and one adjoint electromagnetic simulation. The gradient

information enables systematic updates of the design variables according to the chosen optimization scheme.

Within this gradient-based framework, two principal geometry representations are used to parameterize the design domain: density-based topology optimization and the level-set formulation. Both representations employ the adjoint gradient to modify the material distribution but differ in how the geometry is encoded, how boundaries evolve, and how topological changes are handled. The following subsections describe these optimization methodologies and their respective advantages in nanophotonic device design.

In density-based topology optimization, the design region is discretized into pixels or voxels, each assigned a continuous material-density variable $\rho(r) \in [0, 1]$. The local permittivity is defined through an interpolation relation of the form:

$$\varepsilon_r(\rho) = \varepsilon_{min} + \rho(\varepsilon_{max} - \varepsilon_{min}).$$

This continuous material parameterization enables gradient-based optimization, allowing the permittivity distribution to evolve smoothly during the iterative update process [53].

The adjoint-variable method provides the gradient of the objective function with respect to the density field using only one forward and one adjoint electromagnetic simulation, making the approach computationally efficient for domains containing thousands to millions of degrees of freedom [53]. The continuous representation is compatible with broadband, multi-port, and multi-objective optimization, enabling the discovery of highly compact and non-intuitive device layouts.

The level-set formulation represents the photonic geometry implicitly through a continuous scalar field $\Phi(r)$. The zero-level contour of Φ defines the interface between

different materials, and its evolution follows a Hamilton–Jacobi-type equation driven by the objective function's sensitivity:

$$\frac{\partial \Phi}{\partial t} + V_n|\nabla \Phi| = 0,$$

where V_n denotes the normal velocity derived from the FOM gradient [57].

This representation maintains strictly binary material distribution and produces boundary shapes with controlled curvature and smoothness. The level-set method is therefore well suited for devices requiring refined geometric features and low-scattering transitions, including adiabatic tapers, beam splitters, and mode converters [57].

The method is less adaptable to large topological modifications because newly formed void or solid regions do not automatically emerge unless reinitialization or hybridization strategies are applied. Hybrid approaches have been proposed that combine density-based exploration with level-set refinement or incorporate data-driven surrogate models to accelerate convergence in high-dimensional design landscapes [48]. The optimization strategies in inverse photonic design are founded on adjoint-gradient methods coupled with appropriate geometric representations.

Density-based topology optimization enables extensive topological freedom and compatibility with continuous gradient updates, making it suitable for broadband, multi-functional devices. Level-set optimization provides highly controlled and fabrication-friendly boundary evolution and is advantageous for applications requiring precise shape definition. Together, these methods constitute the core computational framework for solving large-scale inverse-design problems and form the basis for the manufacturing-aware optimization considerations introduced in Section 2.4.

2.4 Manufacturing-Aware Inverse Design and Robustness

Inverse-design optimization often identifies solutions that maximize electromagnetic performance but are not directly manufacturable. These numerically optimal geometries may contain features below lithographic limits, highly oscillatory boundaries, or abrupt index transitions that exceed process capabilities. Imposing manufacturing constraints is therefore necessary, but doing so restricts the allowable design space and inevitably degrades the achievable performance compared to the unconstrained optimum. The goal of manufacturing-aware optimization is to balance performance with geometric realizability and process tolerance.

Minimum feature size is a fundamental manufacturing constraint that limits the smallest reproducible geometric detail in an inverse-designed device. This limit is determined not only by the intrinsic resolution of the fabrication method but also by the device's tolerance to dimensional errors. For example, if an e-beam lithography process exhibits a dimensional uncertainty of approximately 20 nm, selecting a minimum feature size of 200 nm ensures that the printed structure varies only within ±20 nm, corresponding to a relative deviation of about 10%. The acceptable minimum feature size must therefore reflect both the achievable precision of the manufacturing technology and the sensitivity of the device performance to these deviations.

To enforce this constraint during optimization, spatial filtering and geometric convolution operations are applied to the design field so that no features smaller than the prescribed threshold can form. These operations suppress fine-scale variations, eliminate isolated sub-resolution structures, and prevent the emergence of sharp notches or thin

branches that the fabrication process cannot reliably reproduce. The process ensures that the final design is both fabricable and robust under real-world process conditions.

A practical illustration of manufacturing-aware tuning is provided by the inverse-designed multi-dimensional multiplexer in Shen et al. [58]. Fabrication tests revealed that apertures below 100 nm experienced dimensional errors up to ~15 nm, while even 150 nm features deviated by ~7.5 nm due to etching micro-loading effects. Devices fabricated without correction exhibited more than 5 dB additional insertion loss in several TE paths compared with simulation. To compensate for these distortions, the manufacturing calibration method (MCM) reduced designed features by 7.5–15 nm in advance, depending on their size. After applying this tuning, the fabricated device recovered much of its intended performance, achieving up to 6 dB insertion-loss improvement at 1600 nm and restoring a 100 nm operating bandwidth.

Chapter 3: Design and Performance Analysis

3.1 System Overview

To demonstrate the feasibility of an optical backplane interconnect system, we constructed a four-board WDM communication prototype, as illustrated in Figure 3.1. Each motherboard is equipped with a transceiver containing a single transmitter and receiver. During transmission, each board operates on one specific wavelength channel, while the receiver is designed to detect all four wavelengths (850 nm, 880 nm, 910 nm, and 940 nm), enabling multi-channel communication across a shared optical path.

The proposed WDM optical backplane enables scalable link-level data throughput by transmitting multiple wavelength channels simultaneously through a shared photonic routing structure. In typical short-reach optical interconnect systems, individual wavelength channels commonly support data rates of approximately 25 Gb/s per lane, often implemented using directly modulated VCSEL transmitters operating in the 850–940 nm wavelength range. Therefore, a four-channel WDM architecture can achieve an aggregate link bandwidth of approximately 100 Gb/s [6].

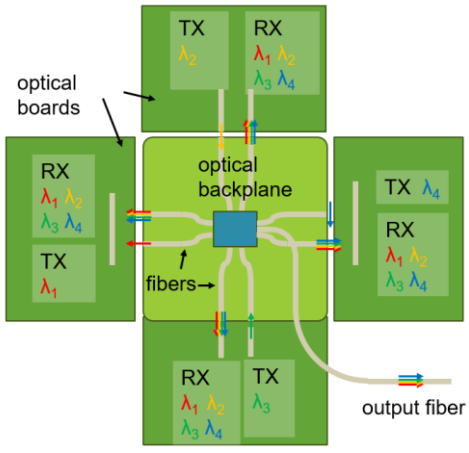


Figure 3.1: Schematic of optical backplane system with four boards.

At the core of the system is an integrated optical chip, highlighted by the blue block in Figure 3.1, which serves as the wavelength-routing hub responsible for aggregating and broadcasting optical signals among the four boards. A schematic of this optical interconnect chip is shown in Figure 3.2. Optical signals from each motherboard are coupled into the chip via on-chip grating couplers, providing efficient vertical coupling from optical fibers (or board-level waveguides) into the planar photonic layer.

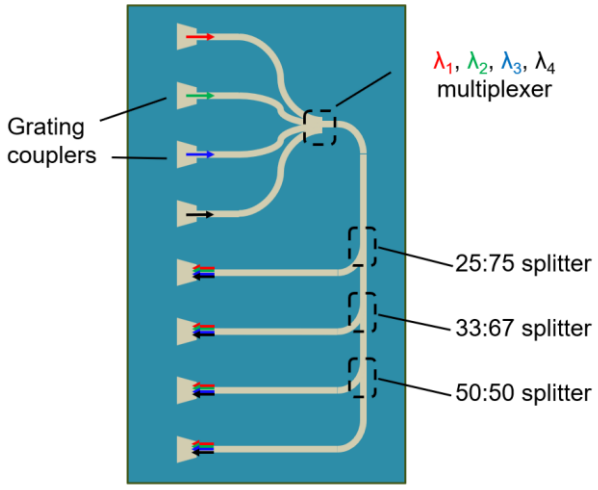


Figure 3.2: Schematic of the optical interconnect chip including grating couplers, aggregating and broadcasting units.

Once inside the chip, the optical signals are routed to the proposed wave-length multiplexing device, a metastructure-based 4×1 multiplexer, which aggregates the four distinct wavelength channels into a single shared optical bus. This bus then feeds into the directional couplers designed for power broadcasting. In this demonstration setup, the directional couplers are designed to evenly split the signal into four output paths, each carrying 25% of the optical power. This thesis does not focus on the entire system design, such as the design of directional couplers and grating couplers. Instead, we will focus more

on the design and performance tuning for the fan-in device, which consists of our four-channel SWDM metastructures.

For the waveguide platform, we selected a silicon nitride (Si_3N_4) core on a silicon-on-insulator (SOI) substrate, as shown in Figure 3.3. This material platform supports operation in the short-wavelength range (λ = 840–950 nm) due to the broad transparency window of Si_3N_4 extending from the visible to the near-infrared regime, whereas crystalline silicon exhibits strong absorption at wavelengths below approximately 1.1 μm [36][37]. In addition, Si_3N_4 waveguides enable low-loss optical signal routing, which is particularly beneficial for broadband wavelength multiplexing and passive power distribution in integrated photonic circuits. Furthermore, the Si_3N_4/SiO_2 material stack remains fully compatible with standard CMOS fabrication processes, making it an attractive choice for scalable photonic integration [37].

To determine the optimal waveguide width, we performed a series of simulations using Finite-Difference Time-Domain (FDTD) methods. These simulations analyzed the effective refractive index (n_{eff}) and propagation loss as a function of waveguide width, across both TE and TM polarization modes at λ = 840 nm and λ = 950 nm. The results are presented in Figure 3.4 (effective index) and Figure 3.5 (propagation loss).

Fig. 3.4 shows that the fundamental TE_{00} and TM_{00} modes remain well confined over a waveguide width range of approximately 300–500 nm, indicating stable guiding behavior without excitation of higher-order modes. The corresponding propagation loss results in Fig. 3.5 further validates this trend, showing that loss remains below 10^{-5} dB/cm within an extended range from roughly 325 nm to 575 nm. Outside this region, loss increases due

to weak confinement or the onset of higher-order modes, leading to leaky or multi-mode behavior. Based on these observations, we conclude that the low-loss, single-mode operating window for both TE and TM polarization spans approximately 350–550 nm. A width within this range was therefore selected for subsequent device design and optimization to ensure efficient guiding, single-mode performance, and fabrication robustness.

It should be emphasized that the extremely low loss values obtained in Figure 3.4 do not represent realistic propagation losses achievable in fabricated silicon nitride waveguides. In eigenmode and FDTD simulations, the estimated loss primarily reflects radiation leakage under idealized material assumptions and finite numerical resolution. In FDTD simulations, numerical accuracy is influenced by factors such as mesh discretization, domain truncation, and boundary conditions, which can lead to underestimation of propagation loss when radiation leakage approaches the numerical noise floor [59]. Therefore, the simulated loss trend is used here as a relative design metric for evaluating modal confinement and determining an appropriate waveguide width range, rather than as an absolute prediction of experimentally achievable propagation loss.

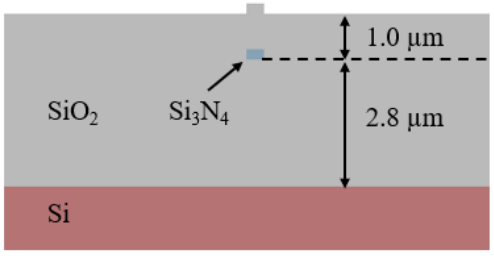


Figure 3.3: Cross-sectional view of the Si_3N_4 waveguide structure on a silicon-on-insulator (SOI) substrate.

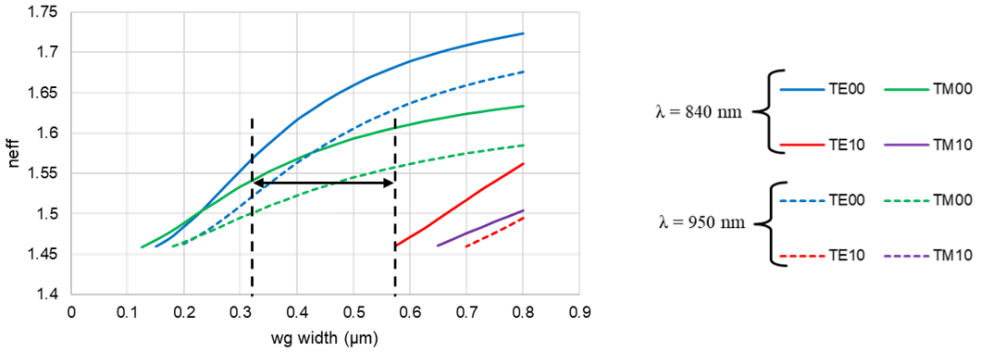


Figure 3.4: Effective refractive index at λ = 840 nm and λ = 950 nm.

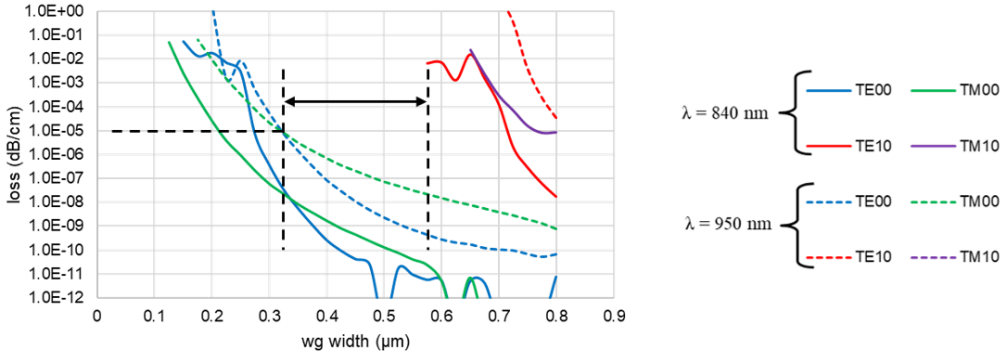


Figure 3.5: Propagation loss versus waveguide width at λ = 840 nm and λ = 950 nm.

Given the need for compact, multi-wavelength routing within this backplane environment, we implemented a photonic device capable of multiplexing and demultiplexing four wavelength channels in a small footprint. This device is integrated at the core of the system to aggregate signals at the transmitter side and to separate them at the receiver side. The detailed design structure and its performance are presented in the following sections.

3.2 Design Parameters and Simulation Setup

The schematic of the proposed four-channel SWDM metastructure is shown in Figure 3.6. The multiplexer features four single-mode waveguide input ports supporting four wavelengths (850nm, 880 nm, 910 nm, 940 nm) and a shared single-mode optical bus waveguide. The fundamental TE mode entering from the input ports is directly routed to the bus waveguide without conversion. The width of the single-mode waveguide is set to 0.5 μm to ensure single-mode propagation and low-loss based on the results in Figure 3.5.

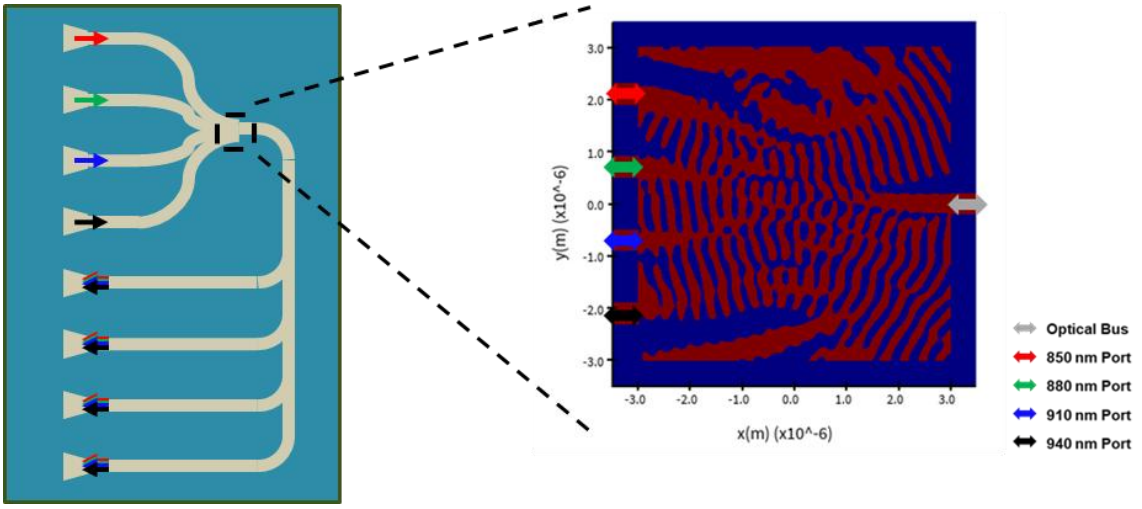


Figure 3.6: The proposed four-channel wavelength-division-multiplexing (WDM) used in the optical backplane.

Our device aims to achieve low insertion loss and effective mode conversion across the entire 100 nm bandwidth within an ultra-compact size of 6 μm × 6 μm. To meet this design goal, we employed a gradient based TO method with inverse design to explore the parameter space by optimizing the permittivity distribution within the design region.

While the proposed design achieves a compact 6 μm × 6 μm footprint, further reduction is constrained by layout and physical limitations. The fan-in configuration

requires ~1 μm waveguide spacing, limiting the available optimization region, while a smaller design area reduces the degrees of freedom for field evolution, degrading performance. In addition, fabrication constraints such as minimum feature size and etch fidelity increase sensitivity to variations, further restricting footprint scaling [20][28][29].

The inverse design workflow adopted in this work consists of four stages: initialization, grayscale optimization, binarization, and design-for-manufacturing (DFM) refinement. In the initialization stage, the figure of merit (FOM), optimization objectives, design domain size, material permittivity range, and other simulation parameters are defined. The structure is represented as a pixel grid, and the initial permittivity distribution (typically uniform or randomized) is assigned.

During the grayscale stage, the optimization region is discretized into pixels, and the permittivity of each pixel is continuously adjusted using gradient-based updates to improve the figure of merit (FOM). Once the optimization converges and the gradient falls below a defined threshold, the structure transitions to the binarization stage, where the permittivity map is quantized into two discrete material states. At this point, each pixel is assigned to either silicon nitride or silicon dioxide, eliminating intermediate values and yielding a physically realizable layout. In the final DFM stage, an indicator function is introduced to penalize FOM contributions from features smaller than the minimum lithographic resolution. This promotes the removal of sub-wavelength holes, pillars, narrow channels, and thin ridges, ultimately guiding the optimizer toward a fabrication-friendly geometry that preserves device performance while ensuring manufacturability.

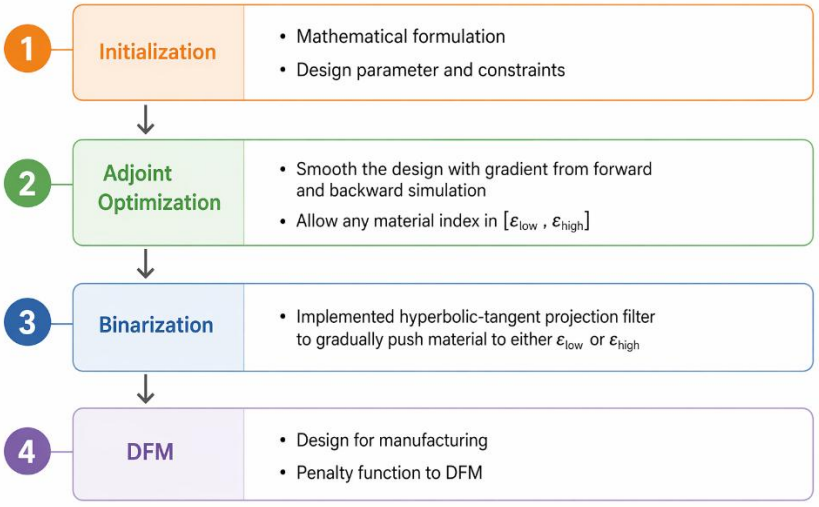


Figure 3.7: Inverse design workflow.

Initialization:

In this work, the inverse-design optimization is driven by a ModeMatch-based figure of merit (FOM), which evaluates how closely the simulated transmission spectrum $T(\lambda)$ matches the target spectral response $T_0(\lambda)$ across the operating wavelength range. The FOM is formulated using an L_p-norm representation over $[\lambda_1, \lambda_2]$, defined as

$$FOM = \left(\frac{1}{\lambda_2 - \lambda_1} \int_{\lambda_1}^{\lambda_2} | T(\lambda) - T_0(\lambda) |^p \, d\lambda \right)^{1/p},$$

which quantifies the deviation between the simulated and target transmission. In this expression, the spectral difference $| T(\lambda) - T_0(\lambda) |$ is raised to the power of p to emphasize mismatch, and subsequently normalized by taking the $1/p$ power, resulting in a smooth and well-behaved metric suitable for gradient-based optimization including adjoint-sensitivity methods and the L-BFGS-B algorithm used in this work. A lower FOM indicates better spectral agreement and higher modal conversion efficiency. In this thesis,

$p = 2$ is chosen, corresponding to an energy-based L_2 norm that penalizes large deviations more strongly while maintaining numerical stability and convergence efficiency. Because FOM integrates over wavelength, the optimization targets broadband performance rather than optimizing at a single wavelength point. The permittivity distribution inside the design region is initialized uniformly, providing a neutral starting point from which the optimizer can iteratively sculpt the refractive-index pattern toward the desired device functionality.

Adjoint Optimization:

The optimization begins in a continuous domain where the design region is pixelated, and each pixel is assigned a permittivity value ranging between the two material states. The objective of this stage is to search the design space freely and identify a permittivity distribution that maximizes the defined figure of merit (FOM), such as transmission efficiency toward the desired output port.

A key enabler in this phase is the adjoint-based gradient optimization method, which allows efficient computation of gradients with respect to all design pixels. In each iteration, two electromagnetic simulations are performed:

- Forward Simulation: The input excitation is launched into the device, and the resulting electromagnetic field distribution is recorded. The FOM is calculated based on the device output (e.g., overlap with target mode).
- Adjoint Simulation: A reverse excitation is launched from the target output port back into the design region. This generates an adjoint field that describes how sensitive the output response is to perturbations inside the device.

By combining the forward and adjoint electric field distributions, the gradient of the FOM with respect to each pixel's permittivity can be calculated using

$$g(r,\lambda) = 2\varepsilon_0 \sum E_i^{fwd}(r,\lambda) E_i^{adj}(r,\lambda),$$

where E_i^{fwd} and E_i^{adj} represent the local forward and adjoint electric fields. Importantly, this formulation computes the gradient for all pixels simultaneously, meaning that whether the design contains hundreds or tens of thousands of pixels, only two simulations per iteration are required. This drastically reduces computational cost and enables fast convergence even in large design spaces.

After gradient computation, the permittivity of each pixel is updated along the direction that improves the FOM, progressively forming an optimized grayscale refractive-index pattern. This stage continues until the improvement in FOM falls below a threshold, indicating that the structure has reached a locally optimal continuous-state solution.

Binarization:

The optimized structure obtained from the grayscale stage contains continuous permittivity values between the two material limits. While this distribution is mathematically optimal for light propagation, it has no direct physical meaning because the fabricated device can only consist of two discrete materials, in our case Si_3N_4 or SiO_2. Therefore, the grayscale result must be converted into a binary permittivity map, where each pixel is assigned exclusively to one material state. This is achieved by applying a smooth thresholding function—commonly implemented using a Heaviside or tanh-based projection filter:

$$\varepsilon_i = \varepsilon_{\text{low}} + \frac{\tanh(\beta\eta) + \tanh[\beta(\rho_i - \eta)]}{\tanh(\beta\eta) + \tanh[\beta(1-\eta)]}\Delta\varepsilon,$$

where ρ_i represents the grayscale density of pixel i, β controls the sharpness of binarization, η is the threshold level, and $\Delta\varepsilon = \varepsilon_{\text{high}} - \varepsilon_{\text{low}}$ defines the material contrast. As β increases throughout optimization, the transition becomes steeper, progressively driving intermediate values toward either ε_{low} (SiO_2) or $\varepsilon_{\text{high}}$ (Si_3N_4).

The objective of this stage is to retain the optical performance achieved during grayscale optimization while converting the solution into a physically fabricable layout. By gradually tightening the projection and controlling the update schedule, abrupt feature collapse is avoided, and crucial field-guiding regions are preserved. The result is a well-defined binary topology with sharp boundaries and no continuous-valued mixtures, preparing the structure for final refinement and fabrication consideration.

Design for Manufacturing:

Following binarization, the structure enters the design refinement stage, where the binary topology is further optimized under practical fabrication considerations. Although binarization removes intermediate permittivity values, the resulting geometry may still contain sub-wavelength holes, narrow channels, or isolated features that are difficult to reproduce in fabrication and highly sensitive to etching variations. To address this, we apply a manufacturing penalty function that suppresses features below the minimum lithographic resolution. This penalty term is incorporated into the FOM, discouraging the formation of excessively small structures and guiding the optimization toward a layout that

maintains optical performance while improving fabrication robustness.

By iteratively updating the binary design in the presence of this penalty, the algorithm smooths fine irregularities, eliminates unstable features, and enhances structural uniformity. The resulting device maintains the performance characteristics achieved during grayscale optimization—such as high coupling efficiency and broadband response—while evolving into a fabrication-tolerant, physically realizable layout. This refined structure is used as the final inverse-designed device for performance evaluation and subsequent implementation in the optical backplane system.

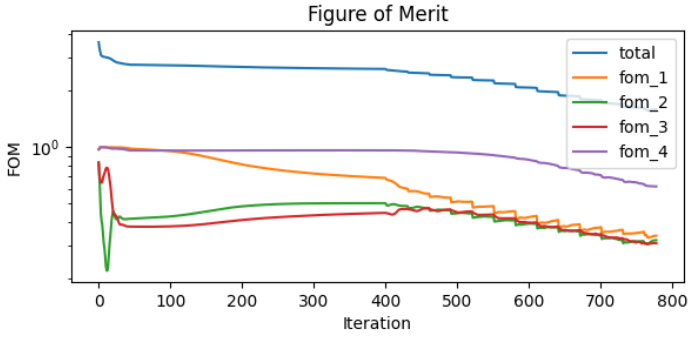


Figure 3.8: The figure of merit plot over iteration.

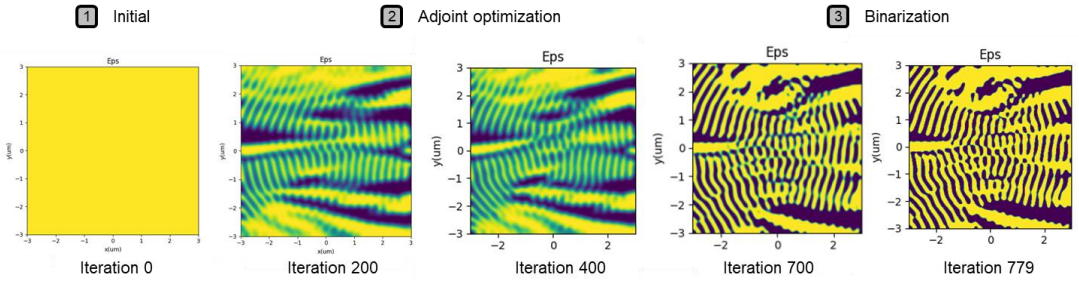


Figure 3.9: The intermediate design structures generated at different optimization steps.

3.3 Results and Performance Evaluation

With the finalized geometry, we proceeded to evaluate the optical performance of the device through three-dimensional FDTD simulations, focusing on wavelength routing behavior and spectral transmission efficiency. Figure 3.10 (a) – (d) shows the simulated electric-field intensity profiles for four individual input wavelengths at 850 nm, 880 nm, 910 nm, and 940 nm. In all cases, the guided light is directed toward the same output bus, with wavelength-dependent modal evolution occurring inside the metastructured region.

The corresponding transmission spectra with no fabrication deviation are presented in Figure 3.10 (e) and Table 3.1. The device exhibits four clear transmission peaks aligned with the design wavelengths, with simulated transmission efficiencies of 65% (850 nm), 72% (880 nm), 75% (910 nm), and 65% (940 nm). These values correspond to insertion losses of approximately -1.87 dB, -1.43 dB, -1.25 dB, and -1.87 dB, respectively. The overall spectral response is smooth and broadband, without narrow resonant features, demonstrating that the inverse-designed metastructures supports robust multiplexing behavior over the 840–950 nm operational range.

All results reported here are obtained from numerical simulation. Fabrication and experimental verification remain planned future work. A 150-nm minimum feature size is enforced during refinement to maintain lithography compatibility with standard I-line or e-beam processes. This fabrication-aware constraint enables straightforward integration into existing Si_3N_4 photonic foundry flows while improving tolerance to pattern deviations.

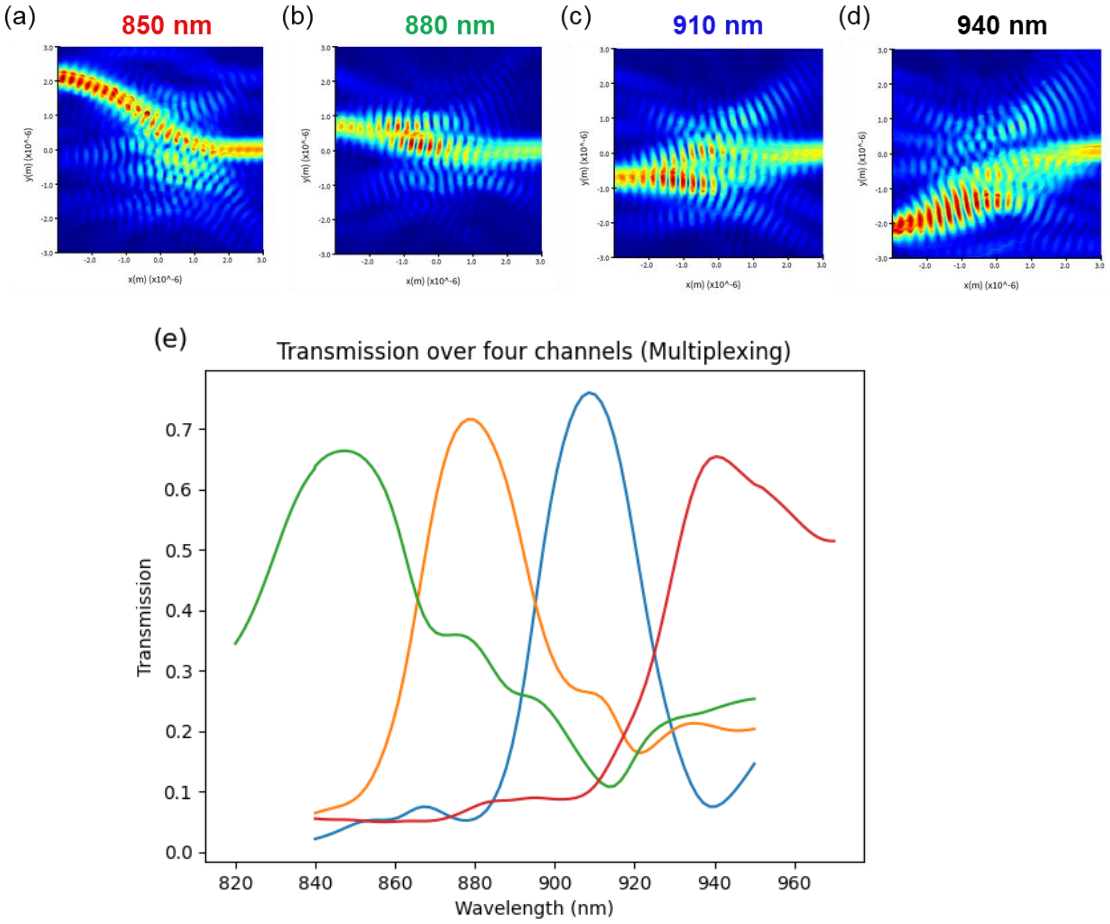


Figure 3.10: Simulated optical performance of the inverse-designed four-channel WDM device. (a–d) Electric-field intensity for individual wavelength channels at 850, 880, 910, and 940 nm. (e) Transmission spectra of the four channels.

While the simulated results demonstrate strong multiplexing performance across the target wavelength range, practical implementation inevitably introduces fabrication-induced variations that can alter the device geometry. In particular, etching processes are known to modify feature dimensions, leading to deviations from the designed structure. Such variations can affect the effective refractive index distribution and, consequently, the

optical response of the metastructure. Therefore, it is important to evaluate the sensitivity of the proposed design to fabrication deviations to ensure reliable operation under realistic conditions.

As shown in Shen et al. [58], fabrication processes such as etching tend to enlarge feature sizes, leading to deviations from the designed geometry. To evaluate the robustness of the proposed structure under such fabrication variations, we systematically increased the size of the etched regions by 5 nm, 10 nm, 15 nm, and 20 nm, and analyzed the corresponding transmission performance.

The impact of fabrication deviation on device performance is further evaluated using the absolute transmission values summarized in Table 3.1 and Figure 3.11. As the deviation increases, all wavelength channels exhibit a gradual reduction in transmission. For the 910 nm channel, the transmission decreases from 75% to 71% and 63% under 5 nm and 10 nm deviations, corresponding to reductions of approximately 5% and 16%, respectively. A more pronounced degradation is observed at larger deviations, where the transmission drops to 50% at 15 nm and further to 36% at 20 nm, representing a total reduction of more than 50% compared to the nominal case. Similar trends are observed across other wavelength channels, indicating consistent sensitivity to geometric variations.

Based on these results, the proposed device can tolerate fabrication deviations of up to approximately 10 nm while maintaining acceptable performance. Beyond this range, the optical response is significantly degraded, indicating that controlling fabrication variation within this tolerance is critical to ensure the device operates as designed.

Fabrication Deviation	Transmission			
	850 nm Channel	880 nm Channel	910 nm Channel	940 nm Channel
0 nm	65%	72%	75%	65%
+5 nm	62%	67%	71%	64%
+10 nm	59%	63%	63%	62%
+15 nm	53%	55%	50%	59%
+20 nm	45%	44%	36%	57%

Table 3.1: Transmission at Different Wavelengths under Various Fabrication Deviations

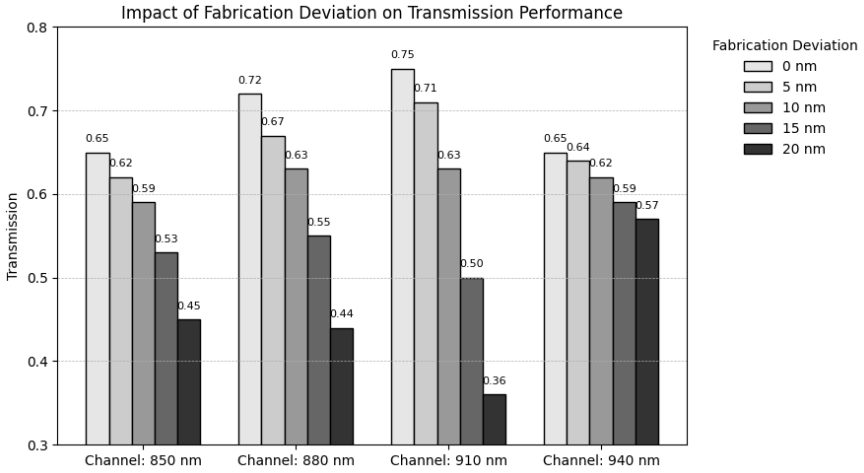


Figure 3.11: Impact of fabrication deviation on transmission performance

Chapter 4: Conclusion

This thesis presents the design and numerical demonstration of an ultra-compact four-channel short-wavelength-division-multiplexing (SWDM) photonic device based on inverse-designed metastructures in a Si_3N_4-on-SiO_2 platform. Motivated by the scaling challenges of electrical backplanes, we explored wavelength-division multiplexing as a means of increasing channel density and reducing link power consumption for future high-speed optical interconnects. By integrating gradient-based topology optimization through an inverse design method, we developed a compact 6×6 μm^2 multiplexing structure capable of routing four optical wavelengths within the 850–940 nm band into a bus waveguide.

A complete optimization workflow was formulated and implemented, consisting of initialization, grayscale optimization, binarization, and fabrication-aware refinement. The optimization was driven by a broadband ModeMatch-based figure of merit, and gradients were efficiently computed using one forward and one adjoint electromagnetic simulation per iteration. Binarization and DFM refinement ensured that the final structure transitioned from a continuous permittivity field to a binary, fabrication-compatible layout while maintaining optical performance.

Three-dimensional FDTD simulations verify that the proposed device achieves high transmission efficiency with simulated peak transmissions of 65% at 850 nm, 72% at 880 nm, 75% at 910 nm, and 62% at 940 nm. The broadband response, smooth spectral profile, and low insertion loss demonstrate the capability of inverse-designed metastructures to realize dense multiplexing within an area far smaller than conventional MMI-based implementations. The final layout satisfies a 150-nm minimum feature size constraint,

enabling compatibility with standard lithography processes and providing a practical path toward fabrication.

Overall, this work confirms that inverse design is a powerful approach to realizing highly integrated photonic elements for next-generation optical backplanes. Future efforts will focus on fabrication of the extracted GDS layout, experimental validation of transmission characteristics, sensitivity analysis under dimensional variation, and system-level integration with the optical backplane demonstrator. The methodologies developed in this work can be extended toward multi-port fan-out structures, polarization-independent components, and larger-scale WDM architectures, offering broad potential for compact, power-efficient photonic interconnect systems.

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Vita

Wei-Cheng Lee was born in Taipei, Taiwan. He earned his Bachelor of Science degree in Electrical Engineering from Purdue University in 2024. In the same year, he entered The University of Texas at Austin to pursue a Master of Science degree in Electrical Engineering, specializing in integrated photonics and optical interconnects.

Address: wl22796@utexas.edu